Le Cao Huu Dang

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Focus

Data Science student (BSc, graduating 2027) building end-to-end AI systems across LLM applications, multimodal AI, APIs, and data pipelines. Experienced in designing modular AI workflows, integrating models with databases and backend services, and deploying containerised applications. Researcher at UIT NLP with hands-on experience in AI system development and technical problem solving; currently expanding cloud architecture skills through AWS.

Experience

Researcher **Dec 2023 – Present**

UIT NLP Research Group • Ho Chi Minh City

- Designed the annotation guidelines and ran the quality-control loop for a Vietnamese Visual Question Answering (ViVQA) dataset(38K images and 50K Vietnamese question–answer pairs), coordinating 20 annotators while contributing labelled samples directly.
- Surveyed Transformer-based VQA architectures to author the related-work section, isolating the fusion-strategy gap that shaped the team's proposed approach.
- Compared candidate architectures and evaluation metrics to decide what the paper would report.

Education

University of Information Technology (VNU-HCM) **2023 – 2027**

BSc Data Science • GPA 3.32/4.0 • Ho Chi Minh City

Technical Skills

Programming: Python, SQL, C++

LLM & AI Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Workflows, Tool / Function Calling, Agent Memory, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM APIs (Groq)

Retrieval & Vector Search: ChromaDB, FAISS, Vector Databases, Semantic Search, Embeddings, sentence-transformers, OpenCLIP

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, Natural Language Processing (NLP), Computer Vision, Speech Recognition (ASR), Machine Translation (MT), Text-to-Speech (TTS), ONNX Runtime, Sherpa-ONNX, Model Quantization (INT8)

Data Engineering: Hadoop/HDFS, Apache Spark (PySpark), ETL / Data Pipelines, Distributed Computing, Big Data Processing, Data Cleaning & Feature Engineering, Exploratory Data Analysis (EDA), Pandas, NumPy

Databases: PostgreSQL (Supabase), InfluxDB, Redis (Caching)

Backend & DevOps: FastAPI, Flask, REST API, WebSocket, asyncio, Node.js (Express), Socket.IO, Docker & Docker Compose, Nginx, Git / GitHub, Linux

Data Visualization: Matplotlib, Seaborn, Recharts

Soft Skills

Soft Skills: Teamwork, Research & Analytical Thinking, Problem Solving, Technical Writing, Communication, presentation skills

English: Working proficiency (written and spoken)

Projects

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[July 2026 – Aug 2026]

- Built an offline speech-to-speech translation pipeline (VAD → ASR → MT → TTS) covering 4 directions (VI↔EN, VI↔ZH) with INT8 ONNX models (Zipformer 30M, Moonshine, Paraformer, Opus-MT/MarianMT, Kokoro), reaching 1.2s - 1.5s voice-to-voice latency and ASR RTF of 0.18–0.2 on i7-7700HQ.
- Implemented a continuous hands-free streaming mode: an adaptive energy-based VAD with a self-updating noise floor (Silero VAD when available) segments speech into utterances, which flow through ASR, MT and TTS workers connected by asyncio queues over a FastAPI WebSocket, with inference offloaded to a ThreadPoolExecutor to keep the event loop free.

- Built a plug-and-play config layer (new ASR/MT/TTS engine or language pair via one config entry) and a web client with a FIFO Web Audio queue that removes overlapping playback; exposed per-stage latency metrics through REST.

Neuro-Symbolic VQA System (SynVQA-UITAIC) | PyTorch, ResNet-50, Transformer seq2seq, Neural Module Networks **[June 2026 – July 2026]**

- Built a three-stage compositional VQA pipeline: a ResNet-50 backbone with learnable positional embeddings producing 49×256 spatial slots, a Transformer seq2seq parser that decodes questions into executable reasoning programs, and a Neural Module Network executor over a 30-class answer vocabulary.
- Trained jointly on answer and program supervision with scheduled sampling ramped 0 to 0.25 to close the teacher-forcing versus greedy-decode gap; AdamW with per-group learning rates (backbone at $0.1\times$) under OneCycleLR.
- Evaluated on the CLEVR and CLOSURE benchmarks using the harmonic mean of the two accuracies to measure balanced compositional generalisation, achieving a score of 0.58.
- Made the executor fault-tolerant to malformed predicted programs via an MLP fallback path, and added full-state auto-resume checkpointing so multi-hour runs survive interruption.

HealthCare QA Copilot (Vietnamese Medical RAG) | FastAPI, LangChain, ChromaDB, Llama 3.1 8B (Groq) **[May 2026 – July 2026]**

- Built a full-stack RAG assistant over 2,200 medical QA pairs curated from 10+ sources including major Vietnamese hospital sites, spanning multiple clinical topics.
- Implemented multi-turn conversation with session-scoped memory and automatic query reformulation that rewrites follow-ups into standalone queries before they reach the vector store.
- Added a domain guardrail that detects out-of-scope questions and declines them politely, so the system never answers where it has no retrieved evidence.
- Served all-MiniLM-L6-v2 embeddings from ChromaDB behind FastAPI; containerised backend, Vite frontend and Nginx into a one-command Docker Compose deployment with Swagger-documented REST endpoints.

YouTube Listening-Behaviour Analytics Platform | Chrome Extension, Node.js, Socket.IO, InfluxDB, Next.js **[Mar 2026 – May 2026]**

- Engineered a three-tier real-time pipeline — Manifest V3 extension hooking the `<video>` element $\rightarrow$ Node.js/Express API $\rightarrow$ InfluxDB v3 plus Supabase — capturing 10,000+ playback events.
- Classified raw playback signals (milliseconds played, watch-duration ratio) into skip, early-skip, replay and completion events, enriched with click-source and time-of-day context.
- Cut YouTube Data API calls with a 24h-TTL Redis cache for category filtering, and pushed live updates to the dashboard over Socket.IO.
- Synced the Supabase JWT from the web app into the extension via cookies for single sign-in across both contexts; surfaced 7-day listening time, skip rate and top tracks in a Next.js and Recharts dashboard.

Distributed Data Pipeline for Cold-Chain IoT Sensors | Hadoop/HDFS, Apache Spark, PySpark **[Sep 2025 – Dec 2025]**

- Provisioned a 3-node Hadoop/HDFS and Spark cluster from scratch — namenode/datanode topology, YARN resource allocation, HDFS replication — and maintained it as shared infrastructure for a 3-person team.
- Built the synthetic data generator producing 43M sensor readings across 142K logistics trips with injected cold-chain failure scenarios, landed into HDFS in a partitioned layout for downstream Spark jobs.
- Implemented distributed preprocessing and feature engineering in PySpark over the 43M-row dataset
- Validated pipeline correctness through large-scale EDA on the generated data before handing curated tables to the modelling track.

Multimodal Video Retrieval System | OpenCLIP EVA02, FAISS, PyTorch, Flask **[Aug 2025 – Sep 2025]**

- Implemented text-to-image and image-to-image keyframe retrieval on OpenCLIP EVA02 embeddings indexed in FAISS, holding top-k latency at 300–500ms over approximately 1M keyframes extracted from 100GB of video across cooking, education, news and lifelog domains (AI Challenge 2025).
- Designed a session-scoped asynchronous pipeline: each upload gets its own workspace while a background worker extracts keyframes, generates embeddings and builds the retrieval index, with the frontend polling processing status.
- Wrote a custom vector-storage module handling embedding and metadata persistence with reload across sessions.

Data Processing & Preparation for Machine Learning | NumPy, Pandas, Seaborn, Matplotlib **[Sep 2024 – Dec 2024]**

- Architected end-to-end data ingestion pipelines to aggregate raw data from external APIs and web sources, organizing outputs into structured storage layers to manage a complete and reproducible data lifecycle.
- Designed rigorous data transformation and validation workflows using Pandas and NumPy, enforcing data integrity and ensuring high-quality, production-ready datasets for downstream ML modeling.

- Conducted comprehensive Exploratory Data Analysis (EDA) with Seaborn to uncover statistical distributions, directly driving robust feature engineering strategies to optimize model readiness.

Hobbies

Music, reading and film.